

$$R = \{A, B, C, D, E, G, F\}$$

$$ABC \rightarrow DE \quad \text{trivial}$$

$$\text{Key: } \{AS\}^+ = \{ABCDEFGF\}$$

$$DE \not\rightarrow F \quad \text{Augmentation Duplicate}$$

$$\text{BCNF? No bc } DE \rightarrow F \quad \uparrow \quad \text{not superkey}$$

$$\cancel{ADE} \not\rightarrow F$$

$$DE \rightarrow G$$

$$G \rightarrow F$$

$$\cancel{ADE} \not\rightarrow F \quad \text{Augmentation } \{DEG\}^+ = \{DEGFG\}$$

$$A \rightarrow B$$

$$B \rightarrow C$$

$$\cancel{A} \not\rightarrow C \quad \text{Transitivity}$$

Normalization using Decomposition

$$\{DEG\}^+ = \{DEFG\} = R_1$$

$$R_2 = (R \setminus R_1) \cup \{DEG\} = \{ABCDEF\}$$

$$\begin{array}{l} R_1 \quad DE \rightarrow G \\ \text{in 2NF} \quad G \rightarrow F \\ \quad \uparrow \\ \quad \text{violates BCNF} \end{array}$$

$$\begin{array}{l} R_2 \quad ABC \rightarrow DE \\ \text{in 2NF} \quad A \rightarrow B \\ \quad \quad B \rightarrow C \quad \leftarrow \text{violates BCNF} \end{array}$$

Decompose R_1 in $R_{1.1}$ and $R_{1.2}$

$$R_{1.1} = \{G\}^+ = \{GF\} \quad \text{in BCNF (G is superkey)}$$

$$R_{1.2} = \{DEG\} \quad \text{in BCNF (only FD is } DE \rightarrow G, \text{ DE is superkey)}$$

$$R_{1.1} \quad G \rightarrow F$$

$$R_{1.2} \quad DE \rightarrow G$$

$R_{2.1} = \{B\}^+ = \{B, C\}$ in BCNF

[DIS] contd. 11/11/25

$R_{2.2} = \{A, B, D, E\}$ in BCNF

$R_{2.1} \quad B \rightarrow C \quad R_{2.2} \quad A \rightarrow B$

but does not have $ABC \rightarrow DE$ ⚡

→ cannot be decomposed to BCNF



$R = \{ \text{Brand, Name, Type, Engine} \}$

BMW	1e	Combust	X1
BMW	1e	E	E1
VW	Caddy	G	G1
BMW	3e	Combust	X1



→ $R_1 = (\text{Engine Type})$

$R_2 = (\text{Brand Name Engine})$



X1	Combust
E1	E
G1	G

BMW	1e	X1
BMW	1e	E1 E1
BMW	3e	X1
VW	Caddy	G1

↑
foreign key

This split was lossy, meaning no information was lost and the relations can be recombined through the foreign key 'Engine'.

Normalization using Synthesis

$R = \{A, B, C, D, E, F, G\}$

~~$R_1 = \{A, B, C, D, E\}$~~

$ABC \rightarrow DE \rightarrow R_1 = (A, B, C, D, E)$

$DE \rightarrow G \rightarrow R_2 = (D, E, G)$

$G \rightarrow F \rightarrow R_3 = (G, F)$

$A \rightarrow B \rightarrow R_4 = (A, B)$

$B \rightarrow C \rightarrow R_5 = (B, C)$

$R_4 \subseteq R_1 \quad \times$
 $R_5 \subseteq R_1 \quad \times$



Result of Synthesis

→ Still not BCNF conform

$R_1 \text{ 2NF } R_2 \text{ BCNF } R_3 \text{ BCNF}$